Adjoint Boundary Control for a Bi-harmonic Type System-Discrete Form

Tags: #CFD #Numerical-Method

Introduction

In this note, we are going to present an iterative algorithm which satisfies circulation conservation and guarantees the correct distribution of boundary vorticity, without computing a large linear system with dense matrices.

Note: Problem Statement for Stokes Flow

Let us consider the coupled system

$$\begin{cases} \nabla^2\psi = \omega \\ \nabla^2\omega = 0 \end{cases}$$

with Dirichlet and Neumann conditions of ψ. The boundary vorticity ω_{BC} is considered as a control variable u, such that

$$\psi = \psi(\omega, u)$$
$$\omega = \omega(u)$$

We introduce a numerical residual

$$J(\tilde{\psi}, \tilde{\omega}, u) = \frac{1}{2}\sum_{i}^{N_{BC}}(\frac{\partial\tilde{\psi}}{\partial n}|_i - A_i)^2$$

where

- N_{BC} is the boundary cell number,
- $\tilde{\psi}$ and $\tilde{\omega}$ are the numerical approximations of ψ and ω,
- $A_i = A(x_i)$ is the prescribed boundary velocity.

Using this numerical residual, we can convert the initial problem to an optimization problem with discrete PDE constraints.

Optimization Problem

Find the control variable u which minimizes

$$J(\tilde{\psi}, \tilde{\omega}, u) = \frac{1}{2} \sum^{N_{BC}} (\frac{\partial \tilde{\psi}}{\partial n}|_i - A_i)^2$$

with the PDE constraints

$$r(\tilde{\psi}, \tilde{\omega}, u) = B\tilde{\psi} - C(\tilde{\omega}, u)$$
$$s(\tilde{\omega}, u) = D\tilde{\omega} - E(u)$$

Let us consider a Lagrangian such that

$$\mathcal{L}(\tilde{\psi}, \tilde{\omega}, u, p, q) = J(\tilde{\psi}, \tilde{\omega}, u) + p\, r(\tilde{\psi}, \tilde{\omega}, u) + q\, s(\tilde{\omega}, u)$$

where p and q are Lagrange's multipliers, and they can also be considered as the transposes of adjoint variables. We can choose p and q such that

$$\frac{\partial \mathcal{L}}{\partial \tilde{\psi}} = \frac{\partial J}{\partial \tilde{\psi}} + p\frac{\partial r}{\partial \tilde{\psi}} = 0$$
$$\frac{\partial \mathcal{L}}{\partial \tilde{\omega}} = \frac{\partial J}{\partial \tilde{\omega}} + p\frac{\partial r}{\partial \tilde{\omega}} + q\frac{\partial s}{\partial \tilde{\omega}} = 0$$

and we will obtain

$$p\frac{\partial r}{\partial \tilde{\psi}} = -\frac{\partial J}{\partial \tilde{\psi}}$$
$$q\frac{\partial s}{\partial \tilde{\omega}} = -(\frac{\partial J}{\partial \tilde{\omega}} + p\frac{\partial r}{\partial \tilde{\omega}})$$

Therefore

$$\frac{\partial \mathcal{L}}{\partial u} = \frac{dJ}{du} = \frac{\partial J}{\partial u} + p\frac{\partial r}{\partial u} + q\frac{\partial s}{\partial u}$$

which is the gradient of J.

On each boundary point, the update scheme of control variable u is given by the gradient descent method

$$u_i^{(k+1)} = u_i^{(k)} + \alpha(\frac{dJ}{du})^{(k)}, \quad i = 1, \ldots, N_{BC}$$

where α is a relaxation number.

Numerical Scheme

The numerical scheme of adjoint boundary control for Stokes flow can be summarized as the following steps:

1. Provide the initial guess of boundary vorticity.
2. Compute the adjoint variables $p^{(k)}$ and $q^{(k)}$ at each iteration.
3. Calculate the gradient and update the value of boundary vorticity with the gradient descent method.
4. Resolve the Laplace equation of ω and Poisson equation ψ.

5. Iterative the previous steps till the J is small enough.

Thought

This method is interesting, because we do not need to solve optimization problems on each boundary points to assure the correct boundary vorticity distribution. The spatial information of the distribution is sketched by the quadratic form numerical residual. This can be illustrated in the continuous sense

$$J = \int_{\partial\Omega} (\frac{\partial\tilde{\psi}}{\partial n} - A(x))^2 \, d\Gamma$$

To compute the residual, it will start with the local mismatch then calculate the sum. However, if we replace the quadratic form by the linear form such that

$$J = \int_{\partial\Omega} (\frac{\partial\tilde{\psi}}{\partial n} - A(x)) \, d\Gamma$$

then due to the linearity of integral

$$\int_{\partial\Omega} (\frac{\partial\tilde{\psi}}{\partial n} - A(x)) \, d\Gamma = \int_{\partial\Omega} \frac{\partial\tilde{\psi}}{\partial n} \, d\Gamma - \int_{\partial\Omega} A(x) \, d\Gamma = \int_{\partial\Omega} \frac{\partial\tilde{\psi}}{\partial n} \, d\Gamma - \gamma_c$$

where γ_c is the circulation given by the prescribed boundary velocity, the spatial information of the distribution will be lost.

Reference:

- Andrew M. Bradley, PDE-constrained optimization and the adjoint method
- Gibert Strang, Computational Science and Engineering, Wellesley-Cambridge Press